

58.4%

top-level theorem

```
graph TD; A[top-level theorem] --- B[proof step]; A --- C[proof step]; A --- D[proof step];
```

proof step

proof step

proof step

Mizar 60 (ITP 2023) counts *whole top-level theorems and lemmas*
57,897 in MML 1147; premises from
the whole library; 420 CPU s portfolio

60.7%

lemma in a development

```
graph TD; A[lemma in a development] --- B[goal]; A --- C[goal]; A --- D[goal];
```

goal

goal

goal

Sledgehammer on the AFP (CICM 2015) counts *proof goals as they arise*
6,934 goals from 128 theories; 30 s
per prover; union trusted as oracle

Same-looking percentages. Different units, different premise regimes, different budgets.